

1 The photograph shows a polar bear.



(Source: © Burcu Gultekin / Alamy Stock Photo)

Polar bears are adapted to live in the Arctic Circle.

Polar bears have a layer of fat beneath their skin. This fat insulates them from the cold and can be used to supply the polar bear with water (metabolic water).

They have a thick fur coat which also insulates them from the cold.

Polar bears feed on seals. The polar bear sits on the ice and waits for a seal to come to the surface of the water to breathe.

(a) Which row of the table describes the types of adaptation of polar bears?

(1)

	Anatomical	Behavioural	Physiological
☒ A	produces metabolic water	hunting by sitting on ice	thick fur coat
☒ B	hunting by sitting on ice	thick fur coat	produces metabolic water
☒ C	thick fur coat	produces metabolic water	hunting by sitting on ice
☒ D	thick fur coat	hunting by sitting on ice	produces metabolic water

(b) What is the niche of a polar bear?

(1)

- ☐ **A** produces metabolic water
- ☐ **B** the Arctic Circle
- ☐ **C** controls the population size of seals
- ☐ **D** uses the layer of fat beneath the skin

(c) Suggest why the polar bear needs to rely on the store of fat to provide it with water.

(2)

(d) Climate change is resulting in habitat fragmentation. Habitat fragmentation causes groups of polar bears to become separated from one another.

Explain the effect that habitat fragmentation could have on the genetic diversity of polar bears.

(3)

(Total for Question 1 = 7 marks)